
DAWN-Cast: Latent Dynamic Adaptive Wavelet Network for Precipitation Nowcasting

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Abstract

Precipitation nowcasting is predicting rainfall evolution over short horizons from radar observations which is critical for disaster response, aviation, and urban resilience, yet remains challenging due to the inherently multi-scale nature of precipitation fields. Existing deep learning approaches typically process all spatial scales uniformly and apply identical transformations across diverse climatologies, leading to inconsistent and often inaccurate predictions. We propose DAWN-Cast, a Latent Dynamic Adaptive Wavelet Network built on wavelet decomposition that induces a physically consistent scale separation in radar fields. Specifically, low-frequency components preserve a coherent large-scale structure, while high-frequency components capture decorrelated fine-scale variability. DAWN-Cast operates on this decomposition in a compressed latent space through two key components. First, a Wavelet Guided Temporal Modelling (WGTm) block decomposes the latent input via a discrete wavelet transform into physically meaningful subbands and applies a dedicated Frequency Adaptive Temporal (FAT) Block per subband, combining a stable MLP with an adaptive Gabor activation that modulates between smooth linear and oscillatory temporal responses, providing a subband specific inductive bias for diverse weather conditions. Second, the Spectral Refinement Spatio-Temporal (SRST) Block integrates a Fourier-domain module for global structure temporal refinement with parallel grouped convolutions for localized refinement, coupled via channel mixing. Comprehensive experiments across four benchmark datasets and ablation studies confirm the effectiveness and contribution of each component, and show that a scale-separated wavelet structure with subband-specific inductive biases drives consistent improvements.

1 Introduction

Precipitation nowcasting focuses on short term prediction of radar observations, typically from a few minutes to hours. The task involves spatio-temporal estimation of future Weather Radar measurements using a short sequence of current and past frames. Reliable nowcasts are crucial for meteorology and contemporary society [1, 2]. They improve preparedness for heavy rains and flooding, help in efficient traffic and energy management, improve aviation safety, and facilitate smarter urban infrastructure. However, the intrinsically complex and rapidly evolving nature of the atmosphere, along with the increasing requirements for real-time, fine-scale, and wide-area forecasts, makes precipitation nowcasting a challenging spatio-temporal prediction task that has attracted significant attention in both the meteorological and machine learning communities [3].

Related Work. Two broad paradigms address precipitation nowcasting: Numerical Weather Prediction and data-driven deep learning based Neural Weather Prediction. Due to the inherent nature of short term forecasting, the skill of neural weather prediction is higher than traditional numerical methods [4, 3], leading to their widespread adoption. Deep learning methods are further divided into

deterministic and probabilistic frameworks. Deterministic models [2, 5, 6, 7, 8, 9, 10, 11] employ recurrent networks, convolutions, or transformers to produce single-valued forecasts, effectively capturing global storm motion but often producing blurry outputs that underestimate high-intensity echoes at longer lead times. Probabilistic frameworks [12, 13, 14] use generative models to sample from the distribution of plausible futures, recovering fine-grained detail but sometimes at the cost of positional accuracy due to the unconstrained stochasticity of the generation process. Further details about related work are discussed in Appendix D.

Research Gaps. Despite strong progress, existing neural nowcasting approaches exhibit three key limitations. First, most approaches do not explicitly exploit the multiscale structure of precipitation fields, where large-scale organization and fine-scale variability arise from distinct physical processes. Recent work [11] attempts to disentangle amplitude and phase to separately model intensity evolution and motion dynamics; however, such formulations still operate within a unified representation and do not explicitly decompose the signal across spatial scales. Second, models typically apply uniform transformations across diverse climatological regimes, limiting their ability to adapt to varying weather dynamics. Applying uniform transformations across these datasets implicitly assumes a shared atmospheric regime, which is a poor inductive bias. Third, while pixel-space methods are computationally expensive at radar resolution, existing latent-space methods [13] compress radar frames purely for efficiency without exploiting the physical scale structure of the data.

Our Contribution. Addressing these limitations requires a representation that respects the multi-scale structure of precipitation fields and provides the model with physically grounded inductive biases suited to the dynamics it must predict. We first leverage wavelet decomposition to separate radar fields into low-frequency components capturing coherent large-scale structure and high-frequency components representing fine-scale variability (Figure 1). We validate this empirically on four datasets through spatial autocorrelation and spectral analysis (Appendix C). Building on this decomposition, we process each subband with dedicated learnable blocks designed to provide the model with appropriate inductive biases suited to the temporal dynamics of each frequency regime across diverse climatological conditions, rather than applying a single uniform transformation across all scales. This scale-separated processing is performed in a compressed latent space to maintain computational efficiency, while a dedicated refinement block addresses both coarse temporal development and fine spatial consistency in the predictions.

We propose DAWN-Cast, a Latent Dynamic Adaptive Wavelet Network for precipitation nowcasting, which introduces three key contributions:

1. We analyze the precipitation field dynamics from a multiscale perspective using wavelet transforms and empirically validate that the 2D discrete wavelet decomposition induces a physically consistent scale separation: low-frequency values correspond to the large-scale structure and high-frequency values correspond to the fine-scale variability of the precipitation field.
2. We propose a novel architecture DAWN-Cast consisting of two major components. The **Wavelet Guided Temporal Modeling (WGTM)** block assigns a dedicated **Frequency Adaptive Temporal (FAT)** block to each wavelet subband, where a dual-stream Gabor-MLP design provides the model with a scale-specific and climatically-aware inductive bias for better predictions. The **Spectral Refinement Spatio-Temporal (SRST)** block then refines the aggregated predictions through complementary global spectral and local spatial branches.
3. DAWN-Cast demonstrates state-of-the-art performance across four datasets, with extensive experimental analyses validating the complementary roles of each proposed component.

1.1 Problem Statement

We formulate precipitation forecasting as a spatio-temporal prediction task [8, 12]. Given a historical observation sequence of the weather radar reflectivity fields of length T_i , denoted by $X = [X_t]_{t=-T_i+1}^0 \in \mathbb{R}^{T_i \times C' \times H' \times W'}$, where C' is the number of channels at each space-time location. $C' = 1$ when only the reflectivity fields of the weather radar are used for nowcasting [1]. The variables H' and W' represent the spatial resolution of each frame. The goal of precipitation nowcasting is then to predict the most probable future sequence $Y = [Y_t]_{t=1}^{T_o} \in \mathbb{R}^{T_o \times C' \times H' \times W'}$ of length T_o .

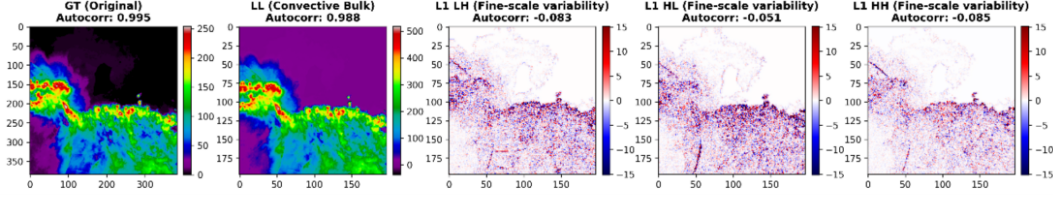


Figure 1: **Spatial scale separation of a convective storm event using Discrete Wavelet Transform (DWT).** Spatial fields of the ground truth precipitation field from SEVIR dataset, alongside its low-frequency (LL) and high-frequency (LH, HL, HH) wavelet subbands. Lag-1 Spatial autocorrelation (referred as Autocorr) values quantify the structural continuity of each field.

2 Methodology

We developed the DAWN-Cast architecture (Fig. 2) for precipitation nowcasting from weather radar images. DAWN-Cast operates in a latent space with two key components (i) Wavelet Guided Temporal Modeling (WGTM) and (ii) Spectral Refinement Spatio-Temporal Block (SRST). In WGTM a wavelet transform factorizes the latent into LL (coherent large scale structures) and HF (fine scale variability) subbands, processes each band with a dual-stream novel Frequency Adaptive Temporal (FAT) block for temporal modeling, later these bands are fused via Inverse wavelet transform. Finally Features are refined with the SRST. More details of each component follows.

Latent Dynamics. We first transform the original radar fields into a compact and structured latent representation using an autoencoder framework that is pre-trained using settings of the LDM model [15]. The encoder $\mathcal{E}$ maps each radar frame from $\mathbb{R}^{C' \times H' \times W'}$ to a latent $z \in \mathbb{R}^{C \times H \times W}$. This latent space serves as a more information-dense domain, where essential precipitation structures are preserved while redundant variations are suppressed. Operating in this space reduces computational complexity and enables the forecasting module to focus on the underlying dynamical evolution rather than pixel-level artifacts [15].

2.1 Wavelet Guided Temporal Modelling (WGTM)

Radar reflectivity fields are inherently multiscale (Fig. 1): large-scale convective structures organize storm morphology over tens of kilometers, while fine-scale turbulent signatures evolve rapidly within localized regions. Processing all spatial scales with a single uniform transformation ignores this physical structure. We address this by first decomposing the latent sequence $\mathbf{Z}$ via a J -level wavelet transform [16], which naturally separates these regimes into a low-frequency approximation subband $\mathbf{S}_J^{LL}$ that encodes coherent large-scale structure of the precipitation field and high-frequency detail subbands $\{\mathbf{S}_j^{LH}, \mathbf{S}_j^{HL}, \mathbf{S}_j^{HH}\}_{j=1}^J$ that captures fine-scale structure of the precipitation field. We validate this physically motivated scale separation empirically in Section 4.1. The complete set of $1 + 3J$ subbands produced by the decomposition is

$$\mathcal{S}_J = \{\mathbf{S}_J^{LL}\} \cup \bigcup_{j=1}^J \{\mathbf{S}_j^{LH}, \mathbf{S}_j^{HL}, \mathbf{S}_j^{HH}\}. \quad (1)$$

A dedicated Frequency Adaptive Temporal (FAT) block is assigned to each subband, allowing the model to learn temporal dynamics that are appropriate to the scale and physical character of each band independently.

2.1.1 Frequency Adaptive Temporal (FAT) Block

This block maps the T_{in} input frames to the T_{out} predicted frames. The central insight behind the FAT block is that different wavelet subbands do not just differ in spatial frequency, they also differ in how these subbands evolve temporally. A low-frequency subband representing a slowly drifting large-scale storm system evolves smoothly across the five input frames. A high-frequency subband representing a rapidly intensifying convective cell changes abruptly and non-linearly across the same five frames. A single shared temporal model cannot be simultaneously well-suited for both. The FAT block addresses this by combining two complementary temporal streams.

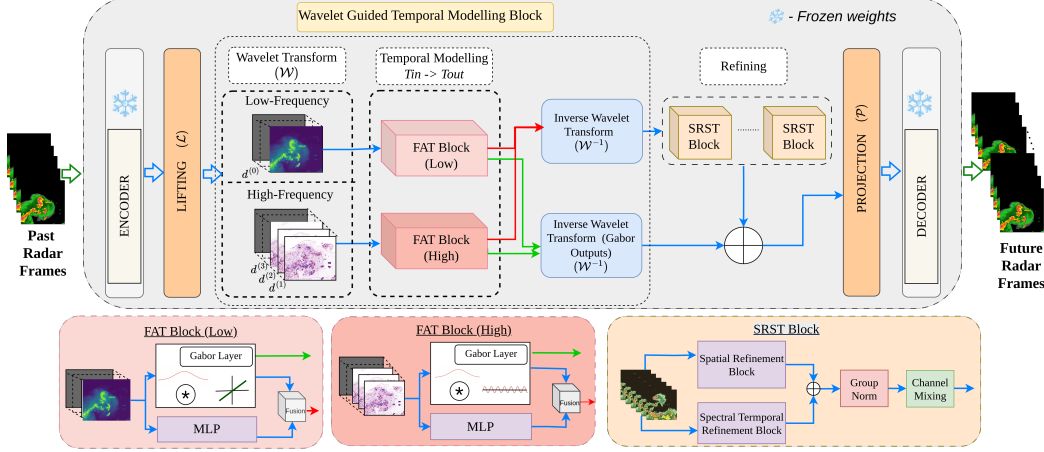


Figure 2: **Overview of the proposed DAWN-Cast architecture.** Past radar frames are encoded and passed through a Lifting ($\mathcal{L}$) operation before the Wavelet Guided Temporal Modelling Block, which decomposes the input into Low-Frequency and High-Frequency components via Wavelet Transform ($\mathcal{W}$). Each component is independently processed by dedicated FAT Blocks (Low and High), followed by Inverse Wavelet Transforms ($\mathcal{W}^{-1}$) to reconstruct the temporal representations, which are then fused via summation. The fused features are further refined through a series of SRST Blocks, each comprising a Spatial Refinement Block and a Spectral Temporal Refinement Block, combined via Group Normalization and Channel Mixing. The refined features are projected ($\mathcal{P}$) and decoded to generate future radar frames. The FAT Block leverages a Gabor Layer alongside an MLP with a Fusion module to effectively capture frequency-aware temporal patterns in both low- and high-frequency components.

127 **MLP stream.** The first stream is a standard MLP applied along the temporal dimension. Its role
 128 is to learn the general nonlinear temporal transformations. It provides a stable, flexible baseline
 129 temporal modeling for every subband.

130 **Gabor stream.** The second stream is the core of the FAT block. The Gabor activation maps are
 131 produced by a learnable sinusoidal filter modulated by a Gaussian envelope,

$$\mathbf{G}_j^k = \sin(\lambda \mathbf{f} \odot (\mathbf{W}^\top \mathbf{X} + \mathbf{b})) \odot \exp(-0.5 \gamma \odot \mathbf{D}(\mathbf{X})), \quad (2)$$

132 where $\mathbf{f}$ and γ are the learnable parameters of the frequency and bandwidth of each neuron, $\mathbf{D}(\mathbf{X})$
 133 is the squared distance from each temporal input to a learned center μ , and $\odot$ denotes elementwise
 134 multiplication. The bandwidth parameters γ are initialized from a $\text{Gamma}(\alpha, \beta)$ distribution to
 135 control the localization effect.

136 What makes this useful is what happens at the extremes of its parameter space. When the effective
 137 frequency λf_j is small and $\gamma_j \approx 0$, the $\sin(z) \approx z$ approximation holds and the Gaussian collapses
 138 to unity, reducing the Gabor neuron to a simple linear projection. When λf_j is large and γ_j is large,
 139 the output oscillates rapidly and is spatially localized around μ_j . Because all of these parameters are
 140 learned end-to-end, the Gabor stream can specialize independently per subband: for the low-frequency
 141 approximation subband $\mathbf{S}_j^{LL}$, where large-scale convective structures may drift slowly or move fast
 142 depending on local dynamic conditions, the model learns what temporal behavior is appropriate; for
 143 the high-frequency detail subbands $\{\mathbf{S}_j^{HH}\}$, where turbulent fine-scale variability can intensify or
 144 dissipate rapidly under strong updrafts, the more sinusoidal or oscillatory regime is more appropriate.
 145 This is the climatic inductive bias, not a hard rule imposed by design, but a parameterized capacity
 146 that the model learns to exploit differently for each subband and each dataset.

147 **Fusion and block output.** The Gabor and MLP streams each produce an intermediate representation
 148 of dimension $C \times H_j \times W_j \times T_{out}$. They are concatenated along the channel dimension and fused
 149 by a $1 \times 1 \times 1$ convolution that projects $2C$ back to C , allowing the model to take a learned weighted
 150 combination of both representations, that is,

$$(\mathbf{G}_j^k, \mathbf{F}_j^k) = \text{FAT}_j^k(\mathbf{S}_j^k), \quad \forall \mathbf{S}_j^k \in \mathcal{S}_J, \quad (3)$$

151 where $\mathbf{G}_j^k$ is the Gabor stream output retained separately and $\mathbf{F}_j^k$ is the fused output.
 152 The $\{\mathbf{F}_j^k\}$ are recombined via the J -level inverse DWT, and the $\{\mathbf{G}_j^k\}$ are recombined via the inverse
 153 DWT to form a separate Gabor residual path as follows

$$\widehat{\mathbf{Z}}_{\text{fused}} = \mathcal{W}_J^{-1} \left(\{\mathbf{F}_j^k\}_{(j,k) \in \mathcal{S}_J} \right) \in \mathbb{R}^{C \times T_{\text{out}} \times H \times W}, \quad (4)$$

$$\mathbf{R}_{\text{Gabor}} = \mathcal{W}_J^{-1} \left(\{\mathbf{G}_j^k\}_{(j,k) \in \mathcal{S}_J} \right) \in \mathbb{R}^{C \times T_{\text{out}} \times H \times W}. \quad (5)$$

154 These recombined outputs are further used by the SRST-Block as discussed in what follows.

155 2.2 Spectral Refinement Spatio-Temporal Block (SRST)

156 The outputs from the WGTM are temporally coherent but spatially raw. The FAT blocks essentially
 157 model each wavelet subband independently along the temporal axis, and the inverse DWT reassembles
 158 them back into a full-resolution latent tensor. What is missing at this stage is joint spatial refinement:
 159 the reconstructed latent needs to be cleaned of artifacts introduced by independent subband processing,
 160 its global temporal structure needs to be sharpened, and its fine-grained spatial details need to be
 161 recovered. The SRST addresses all three of these needs in a single lightweight block through two
 162 parallel refinement streams whose combination is the contribution, not the streams themselves.

163 To enable joint spatio-temporal processing, the temporal and channel dimensions of $\widehat{\mathbf{Z}}_{\text{fused}} \in$
 164 $\mathbb{R}^{C \times T_{\text{out}} \times H \times W}$ are first merged into a single dimension $D = T_{\text{out}} \cdot C$, giving $\mathbf{U} \in \mathbb{R}^{H \times W \times D}$.
 165 This reshaping embeds temporal information into the channel axis so that both spatial and temporal
 166 structure are simultaneously accessible to the refinement branches.

167 **Global structure refinement.** The first stream operates in the Fourier domain. The 2D real
 168 FFT of $\mathbf{U}$ is computed along the spatial dimensions, the resulting spectrum is processed through
 169 two successive complex-valued transformations with grouped block structure (dividing D into N_g
 170 independent blocks of size $d = D/N_g$ for parameter efficiency), and the output is transformed back
 171 via the inverse FFT with a soft-shrinkage regularizer and a residual connection [17]:

$$\mathbf{U}_{\text{spec}} = \mathcal{F}_r^{-1}(\text{SoftShrink}(\Phi(\mathcal{F}_r(\mathbf{U})), \lambda)) + \mathbf{U}, \quad (6)$$

172 where $\Phi(\cdot)$ denotes the two-layer complex-valued frequency-domain transformation and $\lambda > 0$ is the
 173 sparsity threshold. The global receptive field of the Fourier transform is precisely what makes this
 174 branch useful here: unlike convolutions, which only see a local neighbourhood, each output frequency
 175 mode sees the entire spatial field. This makes it effective at suppressing spatially distributed artifacts
 176 and refining the global motion structure of the predicted precipitation field, which is something that
 177 local operations cannot do.

178 **Local fine-detail refinement.** The second stream operates in the spatial domain. A depthwise
 179 convolution with kernel size $k \times k$ is applied independently to each of the D temporal-channel maps:

$$\mathbf{U}_{\text{spa}} = \text{DWConv}_{k \times k}(\mathbf{U}). \quad (7)$$

180 By using groups = D , each channel is filtered by its own independent spatial kernel at a cost of only
 181 $\mathcal{O}(D \cdot k^2)$ parameters. Where the spectral branch refines structure globally, this branch refines it
 182 locally (as shown in Figure6) recovering sharp precipitation boundaries, filling in fine-scale spatial
 183 details, and enforcing local spatial consistency that the Fourier branch is not designed to handle.

184 **Combination and channel mixing.** Neither branch alone is sufficient. The spectral branch has no
 185 spatial locality, whereas the spatial branch has no global context. Their combination is what gives the
 186 SRST its range. The two outputs are summed, normalized, and passed through a pointwise channel
 187 mixing block:

$$\mathbf{O} = \mathbf{cm}(\text{SiLU}(\text{GN}(\mathbf{U}_{\text{spec}} + \mathbf{U}_{\text{spa}}))), \quad (8)$$

188 where $\mathbf{cm}$ indicates channel-mixing. The Group Normalization controls the scale of the combined
 189 signal before it is passed through the channel mixing, which re-mixes information across the merged
 190 temporal-channel dimension effectively performing one final round of lightweight temporal refine-
 191 ment in order to effectively capture the intensity value of precipitation field.

Table 1: Performance comparison of DAWN-Cast with various baseline models on SEVIR, MeteoNet, Shanghai, and CIKM datasets. The best results are highlighted in **bold**, second-best results are underlined, and third-best results are in *italics*. Results of all baselines are obtained from [11].

	SEVIR						MeteoNet					
	CSI-M $\uparrow$	CSI-181 $\uparrow$	CSI-219 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	CSI-24 $\uparrow$	CSI-32 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
ConvGRU	0.2903	0.0879	0.0350	0.3619	0.6100	368.34	0.3401	0.2990	0.1431	0.4667	0.7833	12.85
MAU	0.3076	0.1071	0.0516	0.3863	0.6505	<u>355.48</u>	0.3233	0.2839	0.0997	0.4452	0.7897	12.92
SimVP	0.3108	0.1106	0.0517	0.3924	0.6508	383.56	0.3351	0.3002	0.1130	0.4573	0.7804	13.45
FourCastNet	0.2686	0.0717	0.0339	0.3355	0.5976	410.27	0.3027	0.2533	0.1085	0.4216	0.6450	15.05
EarthFormer	0.2892	0.0844	0.0245	0.3665	0.6633	360.11	0.3205	0.2884	0.1237	0.4491	0.7772	14.43
PhyDNet	0.3017	0.1040	0.0278	0.3812	0.6532	<u>357.63</u>	0.3384	0.3194	0.1366	0.4673	0.7823	14.48
EarthFarSeer	0.3004	0.0992	0.0413	0.3829	0.6327	388.91	0.3404	0.3170	0.1372	0.4726	0.7542	14.10
NowcastNet	0.2791	0.0770	0.0351	0.3512	0.6839	412.94	0.3427	0.3206	0.1598	0.4751	0.7879	15.64
DiffCast	0.3050	0.1300	<i>0.0582</i>	0.3996	0.6482	559.59	0.3512	0.3340	0.1808	0.4846	0.7887	17.93
AlphaPre	<i>0.3259</i>	<i>0.1332</i>	0.0545	<i>0.4110</i>	<i>0.6884</i>	345.18	<u>0.3824</u>	<u>0.3633</u>	<u>0.2002</u>	<u>0.5164</u>	<i>0.7968</i>	<i>12.74</i>
DAWN-Cast(<i>mse</i>)	<u>0.3413</u>	<u>0.1725</u>	<u>0.0937</u>	<u>0.4386</u>	<u>0.7137</u>	394.77	<i>0.3753</i>	<i>0.3552</i>	<i>0.1928</i>	<i>0.5085</i>	<u>0.8300</u>	<u>11.73</u>
DAWN-Cast(<i>faci</i>)	0.3638	0.1950	0.1077	0.4668	0.7284	371.34	0.4085	0.3985	0.2215	0.5482	0.8389	11.54

	Shanghai						CIKM					
	CSI-M $\uparrow$	CSI-35 $\uparrow$	CSI-40 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	CSI-35 $\uparrow$	CSI-40 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
ConvGRU	0.3612	0.3163	0.2062	0.4899	0.7796	33.56	0.3091	0.2009	0.1259	0.4006	0.6507	37.13
MAU	0.3983	0.3621	0.2417	0.5346	0.7195	30.40	0.3039	0.2054	0.1241	0.3928	0.6325	40.74
SimVP	0.3850	0.3549	0.2382	0.5194	0.7795	34.40	0.3052	0.2044	0.1321	0.3955	0.6538	38.06
FourCastNet	0.3571	0.3108	0.2073	0.4868	0.5598	32.10	0.2980	0.1849	0.1015	0.3801	0.4359	<u>36.14</u>
EarthFormer	0.3503	0.3178	0.1872	0.4844	0.7298	35.57	0.3077	0.2039	0.1369	0.4001	0.6267	<i>36.49</i>
PhyDNet	0.3654	0.3236	0.2176	0.4957	0.7751	36.41	0.3038	0.2052	0.1287	0.3931	0.6541	39.56
EarthFarSeer	0.3926	0.3608	0.2343	0.5330	0.5405	32.68	0.3000	0.2046	0.1259	0.3911	0.6373	39.87
NowcastNet	0.3953	0.3608	0.2450	0.5334	0.7902	33.56	0.2991	0.1940	0.1188	0.3865	0.6713	40.96
DiffCast	0.4089	0.3740	0.2606	0.5476	0.7879	36.35	<i>0.3159</i>	0.2009	<u>0.1457</u>	<i>0.4085</i>	0.6499	42.78
AlphaPre	<i>0.4178</i>	<i>0.3854</i>	<i>0.2615</i>	<i>0.5534</i>	<i>0.7951</i>	28.02	<u>0.3194</u>	<i>0.2068</i>	<i>0.1416</i>	<u>0.4137</u>	0.6568	35.18
DAWN-Cast(<i>mse</i>)	0.4368	0.4009	0.2874	0.5792	0.8098	29.01	0.3136	<u>0.2075</u>	0.1278	0.4037	<i>0.6679</i>	37.01
DAWN-Cast(<i>faci</i>)	<u>0.4346</u>	<u>0.3977</u>	<u>0.2758</u>	<u>0.5768</u>	0.8128	29.57	0.3303	0.2349	0.1591	0.4266	<u>0.6696</u>	38.64

192 The block output is reshaped back to $\mathbb{R}^{C \times T_{out} \times H \times W}$ and the Gabor residual $\mathbf{R}_{\text{Gabor}}$ is added,
193 completing the DAWN-Cast module. The predicted latent sequence $\hat{\mathbf{Z}}$ is then decoded by the frozen
194 autoencoder to produce the final T_{out} radar frames.

195 3 Experiments

196 3.1 Experimental Setup

197 **Datasets.** We evaluate our approach on four precipitation datasets. SEVIR [1] covers the United
198 States, providing data at a spatial resolution of $384km \times 384km$ with a 5 minutes temporal resolution.
199 MeteoNet [18] spans a region in France with a coverage of $550km \times 550km$ and a 5 minutes
200 interval between frames. The Shanghai Radar dataset [19] captures precipitation over Shanghai at
201 $501km \times 501km$ with measurements every 6 minutes. CIKM [20] targets the Guangdong region in
202 China, offering data at a $101km \times 101km$ resolution and a 6 minutes temporal interval. Following
203 the configuration of DiffCast and AlphaPre [12, 11], we rescale all images to a spatial resolution of
204 128×128 . For all datasets except CIKM, the input and output frames are set to 5 and 20, respectively.
205 For the CIKM dataset, we use an input length of 5 and an output length of 10. Additional details are
206 provided in the Appendix B.

207 **Baselines.** We compare our proposed model against ten baseline methods: ConvGRU[5], MAU[6],
208 SimVP[7], FourCastNet[21], EarthFormer[8], PhyDNet[9], EarthFarSeer[10], NowcastNet[22],
209 DiffCast[12], and AlphaPre [11] (Previous SOTA). The first five belong to non-disentangled ap-
210 proaches, whereas the remaining four are disentangled methods. Among the latter, DiffCast is a
211 diffusion-based probabilistic model.

212 **Implementation Details.** DAWN-Cast framework is trained on latent representations obtained using
213 a frozen autoencoder [15]. The model is trained for 50 epochs with a batch size of 4 using the
214 AdamW optimizer and a learning rate of 1×10^{-4} . We train two versions of our model, one with
215 MSE Loss and other with Fourier Amplitude and Correlation Loss (FACL) with a constant ratio of
216 0.1 [23] which gives two different versions of our model DAWN-Cast(*mse*) and DAWN-Cast(*faci*).
217 All experiments are conducted on a single NVIDIA RTX A6000 GPU (48 GB VRAM). Additional
218 implementation details are provided in Appendix F.

219 **Metrics.** To evaluate model performance, we employ four widely used metrics Critical Success
220 Index (CSI-M), Heidke Skill Score (HSS), Structural Similarity Index Measure (SSIM), and MSE to

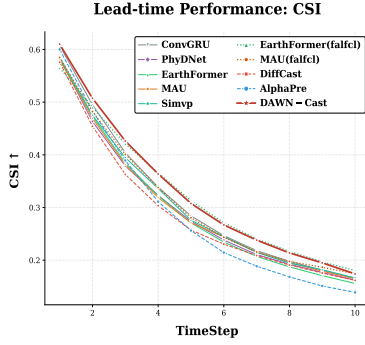


Figure 3: Lead time performance of DAWN-Cast vs baselines on the CIKM dataset.

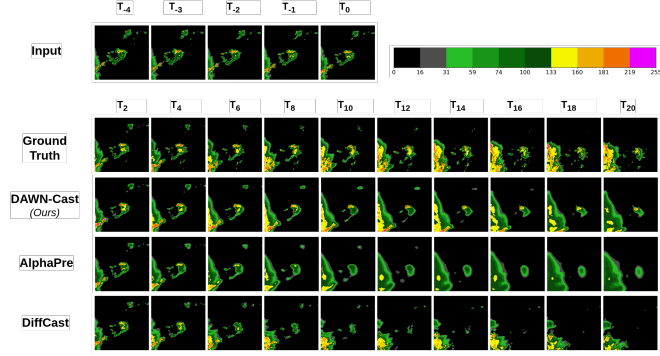


Figure 4: Qualitative comparison of DAWN-Cast and the best performing baselines on SEVIR. T_i denotes temporal offset from T_0 .

221 assess prediction quality, following [8, 12]. The CSI quantifies how accurately the model predicts
 222 precipitation events, considering both false alarms and missed events. In line with [8, 12], we compute
 223 multiple CSI scores. We then report the mean CSI (CSI-M) as an aggregate measure of overall
 224 accuracy. The HSS measures how much the model’s predictions improve upon random guessing,
 225 taking into account both correct and incorrect classifications.

226 3.2 Main Results

227 3.2.1 Quantitative Results.

228 Table 1 presents the performance comparison of DAWN-Cast against ten baseline models in four
 229 benchmark datasets: SEVIR, MeteoNet, Shanghai, and CIKM. DAWN-Cast trained with MSE loss
 230 demonstrates superior performance over all baselines on the majority metrics. Specifically, on the
 231 SEVIR dataset, DAWN-Cast achieves the best CSI-M of 0.3413, CSI-181 of 0.1725 and CSI-219
 232 of 0.0937, while achieving the lowest MSE of 394.77. On MeteoNet, it achieves the best MSE of
 233 11.73, outperforming the strongest baseline AlphaPre (12.74). Similarly, on the Shanghai dataset,
 234 DAWN-Cast(mse) achieves the best CSI-M of 0.4368 and HSS of 0.5792 with a competitive MSE
 235 of 29.01. On the CIKM dataset, it achieves strong results across CSI and HSS metrics. Further,
 236 replacing MSE with FACL loss (DAWN-Cast(facl)) consistently improves performance across nearly
 237 all metrics and datasets. On SEVIR, it achieves the best CSI-M of 0.3638, CSI-181 of 0.1950, HSS
 238 of 0.4668, and SSIM of 0.7284. On MeteoNet, DAWN-Cast(facl) attains the best MSE of 11.54,
 239 CSI-M of 0.4085, and SSIM of 0.8389. These results demonstrate that FACL loss further refines
 240 the model’s ability to capture fine-grained spatiotemporal patterns, establishing DAWN-Cast as the
 241 state-of-the-art across all four benchmarks.

242 We also analyze the performance of the model as a function of the prediction horizon in Fig. 3. DAWN-
 243 Cast maintains consistently strong performance across all lead times for CSI metrics, indicating
 244 stable temporal modeling without rapid degradation. In particular, compared to the baseline methods,
 245 DAWN-Cast exhibits a slower decline in both CSI and HSS at longer lead times, suggesting improved
 246 robustness.

247 3.2.2 Qualitative Results.

248 Figure 4 illustrates sample forecast for the SEVIR dataset, showing that other models typically
 249 underestimate or overestimate precipitation intensity at later lead times, while our model produces
 250 higher and more accurate intensity estimates. Qualitatively, our model better preserves structure and
 251 intensity regions compared to the baselines, in agreement with the quantitative analysis in Section
 252 3.2.1. Additional Qualitative results showing plots for incremental component addition and plots for
 253 other datasets are provided in Appendix H.

4 Model Analysis

DAWN-Cast is built on three claims: that wavelet decomposition provides a physically meaningful scale separation in radar fields, that subband-specific Gabor-MLP temporal modeling provides a useful climatic inductive bias, and that the SRST Block provides necessary spatial-temporal refinement. The following subsections validate each claim in turn through targeted experiments and analyses, establishing that every design decision is both necessary and non-redundant.

4.1 Wavelet Analysis

We claimed that discrete wavelet decomposition induces a physically consistent scale separation in precipitation fields. We validate this through two complementary analyses and a targeted ablation.

Spatial autocorrelation and spectral analysis. Figure 1 provides a qualitative illustration on a single storm event: the LL subband retains coherent large-scale structure ($\rho = 0.988$) while high-frequency subbands (LH, HL, HH) are spatially decorrelated ($\rho \approx -0.06$ to -0.08). To establish this as a dataset-wide property rather than a single-instance observation, Figure 5 in Appendix, presents ensemble autocorrelation and radial PSDs averaged across all events in all four datasets. The LL subband consistently maintains $\rho > 0.95$ across all climatologies while high-frequency subbands universally fall below zero, confirming structural continuity in the low-frequency regime and decorrelated turbulent variability in the high-frequency regime regardless of geographic region or precipitation type. The PSDs reinforce this: the LL subband (blue) dominates at low wavenumbers and closely tracks the ground truth (black), confirming its role corresponding to large-scale convective structure, while high-frequency subbands (red) retain significant energy at high wavenumbers consistent with fine-scale variability decoupled from the macro-scale convective flow.

Ablation. To confirm that both frequency regimes carry non-redundant information, we remove temporal modelling from each regime independently and report results in Table 2. Removing low-frequency temporal modelling causes a large drop in CSI-M (e.g., CSI-M 0.3303 $\rightarrow$ 0.2971 on CIKM, 0.4085 $\rightarrow$ 0.3801 on MeteoNet) and SSIM, reflecting the dominant role of large-scale structure in prediction skill and structure. Removing high-frequency temporal modelling also degrades performance consistently (CSI-M 0.3196 on CIKM, 0.3879 on MeteoNet), confirming that fine-scale dynamics contribute independently. Together, these results justify the dedicated per-subband FAT block design of the WGTM. Wavelet type and level ablations are given in the Appendix E.1.

4.2 Component Analysis

Component ablation. Table 2 reports the contribution of each DAWN-Cast component by removing it individually. Removing the full SRST Block causes the largest drop in metrics on SEVIR data (CSI-M 0.3638 $\rightarrow$ 0.3302), confirming that spatial-temporal refinement is critical after wavelet-guided temporal modeling. Removing the wavelet transform degrades performance across both datasets (SEVIR CSI-M 0.3591, MeteoNet 0.4024), validating that subband-specific processing is necessary and a single uniform temporal transformation is insufficient. Removing the Gabor stream, spectral branch, spatial branch, and channel mixing each causes consistent drops, demonstrating that every component plays a non-redundant role. Qualitative incremental visualizations in the Appendix H further show how prediction quality progressively improves as each component is added.

Adaptive climatic inductive bias. Table 3 confirms that learnable Gabor parameters consistently outperform fixed ones, demonstrating that end-to-end learning allows the FAT Block to self-calibrate its temporal inductive bias to each dataset’s atmospheric character. Table 4 provides the sharpest validation of this: on SEVIR Storm events which are dominated by rapidly evolving convective systems, the strongly sinusoidal Gabor initialization (Config B) outperforms the near-linear one (Config A). At the same time, on SEVIR Random events that contain a comparatively slow and smooth precipitation evolution, the ordering reverses. This behavioral reversal directly mirrors the theoretical premise of Section 2.1.1: the Gabor activation provides the right inductive bias for the dynamical regime it encounters, specializing towards oscillatory responses when precipitation evolves rapidly and towards smooth linear responses when it does not.

Table 2: Ablation study of DAWN-Cast on SEVIR and MeteoNet. Each row removes or replaces one component. *w/o Low/High frequency temporal (WLFT/WHFT)*: substitutes the last observed low/high-frequency wavelet frame across all predicted timesteps, isolating the contribution of each frequency band to temporal modelling. Remaining rows ablate individual components. Best results in **bold**.

	SEVIR						MeteoNet					
	CSI-M $\uparrow$	CSI-181 $\uparrow$	CSI-219 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	CSI-24 $\uparrow$	CSI-32 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
w/o Wavelet transform	0.3591	0.1880	0.1056	0.4598	0.7280	362.99	0.4024	0.3901	0.2204	0.5404	0.8393	11.26
w/o WLFT	0.3460	0.1683	0.0907	0.4410	0.7237	372.03	0.3801	0.3588	0.1970	0.5137	0.8356	11.68
w/o WHFT	0.3409	0.1694	0.0924	0.4396	0.7143	404.54	0.3879	0.3758	0.2048	0.5256	0.8312	12.46
w/o Gabor	0.3541	0.1803	0.0994	0.4532	0.7218	366.26	0.4002	0.3896	0.2129	0.5391	0.8386	11.69
w/o Spectral Branch	0.3526	0.1806	0.0995	0.4520	0.7228	377.77	0.3920	0.3778	0.2095	0.5299	0.8354	11.77
w/o Spatial Branch	0.3552	0.1809	0.0997	0.4550	0.7281	370.27	0.3945	0.3824	0.1970	0.5331	0.8411	11.05
w/o Channel-Mixing	0.3381	0.1418	0.0717	0.4275	0.7262	365.39	0.3998	0.3861	0.2124	0.5382	0.8363	11.41
w/o SRST Block	0.3302	0.1575	0.0787	0.4256	0.6948	413.86	0.3701	0.3543	0.1894	0.5082	0.8236	12.56
w/o WGTB-Block	0.3566	0.1789	0.0979	0.4551	0.7206	360.20	0.4054	0.3941	0.2155	0.5434	0.8402	11.25
Ours	0.3638	0.1950	0.1077	0.4668	0.7284	371.34	0.4085	0.3985	0.2215	0.5482	0.8389	11.54

Table 3: Learnable(Adaptive) Gabor vs Non-learnable (Non Adaptive Gabor)

	SEVIR						CIKM					
	CSI-M $\uparrow$	CSI-181 $\uparrow$	CSI-219 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	CSI-35 $\uparrow$	CSI-40 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
Non-Adaptive Gabor	0.3590	0.1866	0.1006	0.4594	0.7265	370.64	0.3294	0.2292	0.1529	0.4239	0.6586	38.91
Adaptive Gabor	0.3638	0.1950	0.1077	0.4668	0.7284	371.34	0.3303	0.2349	0.1591	0.4266	0.6696	38.64

Table 4: Validation of the FAT Block’s climatic inductive bias under contrasting atmospheric dynamics. SEVIR Storm (rapidly evolving convection) and SEVIR Random (slowly evolving precipitation) are evaluated under Config A (near-linear Gabor) and Config B (strongly sinusoidal Gabor).

	SEVIR Storm			SEVIR Random		
	CSI $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	CSI $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$
Config A	0.3017	0.3568	0.6467	0.3147	0.4053	0.7114
Config B	0.3060	0.3598	0.6392	0.3071	0.3973	0.7124

4.3 Loss and Latent Space Contribution

Table 1 shows that the Fourier Amplitude and Correlation Loss (FACL)[23] consistently improves over the MSE baseline across the majority of datasets indicating the importance of loss function. Importantly, DAWN-Cast(mse) itself performs better compared to other baselines on the majority of metrics, demonstrating that the performance gains of our architecture are robust and not dependent on the choice of loss function. Latent-space operation further contributes mainly to efficiency: operating on compressed representations reduces computational cost.

5 Conclusion and Future work

We presented DAWN-Cast, a latent-space precipitation nowcasting framework built on the observation that wavelet decomposition induces a physically consistent scale separation in radar reflectivity fields. By processing each wavelet subband with a dedicated FAT Block, whose Gabor-MLP dual stream provides a climatically grounded temporal inductive bias and refining the aggregated predictions through the SRST Block, DAWN-Cast achieves state-of-the-art performance across four benchmark datasets, with the largest gains at high-intensity precipitation thresholds where accurate nowcasting matters most.

Future work will explore: (i) a physics-informed autoencoder trained specifically for precipitation dynamics, replacing the general-purpose encoder; (ii) scaling the adaptive part to a larger fraction of model capacity to assess its full potential; and (iii) extending the wavelet-guided temporal modeling paradigm to broader spatiotemporal forecasting tasks, especially in the field of computer vision.

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A Limitations

(i) The learnable parameters of the FAT blocks constitute approximately minor percentage of the total model parameter count. Consequently, the per-dataset ablation score deltas attributable to the adaptive component alone are modest. We interpret this not as a weakness but as a property of the design: the FAT block provides a small-parameter, physically-motivated inductive bias that demonstrably specializes to local climatology without harming overall performance, as evidenced by the consistent subband-specific frequency specialization observed in the section 4.2. (ii) Reliance on a pretrained per-dataset autoencoder transfers some compute cost off-line and may limit flexibility for cross-dataset generalization. (iii) Wavelet decomposition is fixed for a particular dataset; learned wavelets are an open direction to be explore in this field.

B Additional Dataset Details

SEVIR dataset [1] is a curated, labeled dataset that synchronizes several data modalities, including visible satellite imagery, infrared satellite imagery (mid-level water vapor and clean longwave window channels), NEXRAD radar mosaics of vertically integrated liquid (VIL), and ground-based lightning detections. It contains 20,393 distinct weather events recorded between 2017 and 2020, with each event consisting of a 4-hour sequence of images sampled every 5 minutes over a $384 \text{ km} \times 384 \text{ km}$ region within the continental U.S. For the task of forecasting future VIL frames up to 20 time steps (100 minutes) from 5 input frames (25 minutes), we extract sequences of 25 consecutive frames using a stride of 13 for each event. The dataset is partitioned into training, validation, and test subsets based on temporal cutoffs of January 1, 2019, and June 1, 2019. All frames are linearly rescaled to the range 0–255 and then binarized using the thresholds [16, 74, 133, 160, 181, 219] to compute CSI and HSS.

METEONET dataset [18] is a multimodal dataset that includes complete time series of satellite and radar imagery, numerical weather prediction outputs, and ground-based measurements. It covers a $550 \text{ km} \times 550 \text{ km}$ region in northwestern France and spans a three-year period (2016–2018), with data recorded every 5 minutes. Adopting the same protocol as SEVIR, we follow [12] and [11] to partition the radar sequences collected from 2016 to 2018. For the CSI and HSS metrics, we adopt thresholds of [12, 18, 24, 32] following [12] [11]. Detailed dataset statistics are summarized in Table 5.

Shanghai Radar dataset [19] comprises continuous radar echo frames produced by volumetric scans conducted roughly every 6 minutes between October 2015 and July 2018 in Pudong, Shanghai. Each radar reflectivity map spans an area of $501 \text{ km} \times 501 \text{ km}$. We preprocess the echo sequences following the procedures in [12] [11]. The pixel value range of the frames is constrained to [0–70], and thresholds of [20, 30, 35, 40] are used for calculating CSI and HSS.

CIKM dataset [20] originates from the CIKM AnalytiCup 2017 Competition and captures precipitation events over a $101 \text{ km} \times 101 \text{ km}$ region in Guangdong, China. Each data sample consists of 15 past radar echo frames, with a 6-minute interval between successive frames. We followed the preprocessing protocol of [11] [12], we pad each radar echo frame to 128×128 pixels and partition the dataset into training, validation, and test subsets in accordance with the original configuration. Pixel intensities in each frame are converted to reflectivity values in the range [0, 76] dBZ, and thresholds of [20, 30, 35, 40] are employed to calculate CSI and HSS. Detailed dataset statistics are summarized in Table 5.

Table 5: Data split statistics. N_{train} , N_{valid} and N_{test} refer to the number of samples allocated to the training, validation, and test splits.

Dataset	N_{train}	N_{valid}	N_{test}	(C, H, W)	T_i	T_o
Shanghai	1534	526	526	(1,128,128)	5	20
MeteoNet	6308	1310	1310	(1,128,128)	5	20
CIKM	8000	2000	4000	(1,128,128)	5	10
SEVIR	23808	6016	8100	(1,128,128)	5	20

C Wavelet Interpretability

Figure 5 presents a dataset-wide statistical validation of the wavelet scale separation underlying DAWN-Cast. For each of the four datasets: SEVIR, MeteoNet, CIKM, and Shanghai, we compute the lag-1 spatial autocorrelation ρ and the mean normalized radial Power Spectral Density (PSD) across all precipitation events, reporting ensemble statistics over the full test split.

Spatial autocorrelation (top row). The boxplots reveal a consistent and dataset-independent pattern. The ground truth (GT) and the LL subband both exhibit high spatial autocorrelation ($\rho > 0.95$ in all four datasets), confirming that the low-frequency approximation faithfully preserves the coherent large-scale structure of the precipitation field. In contrast, the high-frequency detail subbands (LH, HL, HH) universally exhibit near-zero or negative autocorrelation ($\rho \leq 0$), characteristic of spatially decorrelated signals. This decorrelation is the statistical signature of turbulent fine-scale variability structures that change rapidly across space and have no persistent large-scale organization. The interquartile ranges of the LL subband remain narrow and well above zero in all cases, while those of the high-frequency subbands remain tightly clustered around zero, indicating that this scale separation is not only consistent on average but also stable across individual precipitation events.

Normalized Power Spectral Density (bottom row). The radial PSDs provide a complementary spectral view of the same separation. The LL subband (blue) concentrates its energy at low wavenumbers and closely follows the ground truth spectrum (black) across the entire low-wavenumber regime, confirming that the approximation subband captures the dominant large-scale spectral content of the precipitation field. The high-frequency subbands (red) exhibit a markedly different spectral profile: their energy is concentrated at high wavenumbers and decays steeply, with a significantly shallower slope than the LL subband corresponding to fine-scale variability. The shaded regions (± 1 standard deviation across events) are narrow for the LL subband and wider for the high-frequency subbands, reflecting the greater event-to-event variability in variable fine-scale structure compared to the more predictable large-scale convective organization. Together, the autocorrelation and PSD analyses confirm that 2D discrete wavelet decomposition induces a physically meaningful scale separation in precipitation fields, providing the empirical foundation for the subband-specific temporal modeling design of the WGTM block.

D Related Work

Precipitation Nowcasting with Deep Learning. Precipitation nowcasting focuses on forecasting the near-term evolution of radar observations and has witnessed rapid progress with deep learning. Early approaches such as ConvLSTM [2] introduced convolutional recurrent architectures to simultaneously capture spatial correlations and temporal dynamics, establishing the basis for numerous CNN–RNN hybrid models [24, 25, 26, 27, 28, 5]. Later approaches, including PredRNN [24], refined memory designs to more effectively model long-range spatiotemporal interactions, while PhyDNet [9] integrated physics-inspired constraints to enforce more realistic dynamics. More recently, recurrent-free encoder–decoder architectures like SimVP [7] and transformer-based models such as Earthformer [8] and MIMO [29] have shown that global spatiotemporal dependencies can be captured successfully without relying on explicit recurrence.

Despite these advances, many deep learning approaches still have difficulty producing sharp, realistic radar echoes; their predictions are frequently blurred because regression-based loss functions tend to average over possible outcomes. To overcome this issue, adversarial training schemes have been proposed, as in DGMR [30] and NowcastNet [22], where spatial–temporal discriminators drive the generated outputs toward the true precipitation distribution. Yet, adversarial methods can suffer from unstable training dynamics and visual artifacts. More recently, diffusion-based approaches [31, 32, 33, 34], including DiffCast [12], have been introduced to enhance generative fidelity and more accurately represent uncertainty in precipitation evolution. Prediff [14] further proposes a conditional latent diffusion framework tailored to probabilistic forecasting and integrates a knowledge control component to enforce domain-specific physical constraints on the predictions. However, diffusion models typically demand substantial computational resources because of their iterative sampling procedures [35] and can be highly sensitive to architectural choices, which restricts their practicality [36].

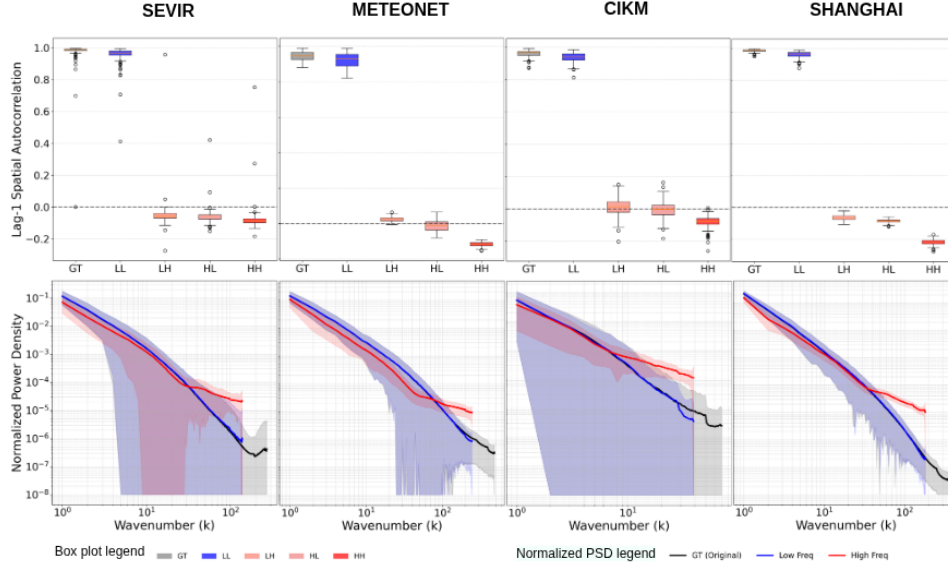


Figure 5: Statistical validation of wavelet-based convective scale separation across diverse meteorological datasets. Ensemble analysis of spatial autocorrelation (*top row*) and mean normalized Power Spectral Density (*bottom row*) across precipitation events from SEVIR, CIKM, Shanghai, and MeteoNet. Boxplots demonstrate that the LL subband consistently maintains structural continuity ($\rho > 0.95$) across all climatologies, whereas high-frequency subbands universally exhibit rapid decorrelation ($\rho \leq 0$) characteristic of turbulence. Here ρ indicates lag-1 spatial autocorrelation. The ensemble PSDs (shaded regions represent ± 1 standard deviation) confirm a robust, dataset-independent decoupling of macro-scale convective flows from localized high-frequency perturbations.

To address issues such as blurry predictions, unstable generative training, and poor scalability, it is natural to consider models that directly capture the dynamical operators underlying precipitation evolution. Neural operators [37] provide this framework by learning continuous mappings between input and output fields, which makes them particularly suitable for spatiotemporal forecasting problems. Notably, neural operators have already demonstrated significant advantages in solving and approximating solutions of partial differential equations (PDEs) [38, 39, 40, 41, 42, 43, 44, 45], where they efficiently learn complex dynamics across varying initial conditions and resolutions, highlighting their strong potential for modeling physically governed atmospheric processes.

Despite their effectiveness, these approaches primarily focus on large-scale atmospheric evolution or maintaining physical consistency, and they do not explicitly represent the localized, high-frequency convective patterns that are crucial for generating extreme precipitation events. To overcome this limitation, we propose DAWN-Cast.

E Additional Ablations

E.1 Wavelet ablations

Tables 6 and 7 show that DAWN-Cast is robust to the choice of wavelet family and decomposition level: performance varies by less than 0.003 CSI-M across wavelet types (Haar, db4, db6) and less than 0.005 CSI-M across levels (1–4) on all four datasets. This insensitivity confirms that the physical scale separation induced by the DWT is a general property of radar reflectivity fields and not an artifact of a specific wavelet basis. We select db6 at level $J = 2$ for SEVIR and $J = 1$ for the remaining datasets based on validation performance; the full type-level grid search is reported in Table 8.

Table 6: Ablation study of DAWN-Cast with various Wavelet types

	SEVIR				MeteoNet			
	CSI-M $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
Haar	0.3520	0.4494	0.7264	372.49	0.4047	0.5443	0.8385	11.45
db4	0.3603	0.4619	0.7289	367.03	0.4061	0.5454	0.8391	11.57
db6	0.3638	0.4668	0.7284	371.34	0.4085	0.5482	0.8389	11.54

	Shanghai				CIKM			
	CSI-M $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
Haar	0.4226	0.5651	0.8118	29.78	0.3291	0.4252	0.6767	37.16
db4	0.4281	0.5701	0.8089	30.79	0.3303	0.4266	0.6696	38.64
db6	0.4345	0.5768	0.8128	29.57	0.3273	0.4228	0.6748	38.71

Table 7: Ablation study of DAWN-Cast with various Wavelet levels

	SEVIR				MeteoNet			
	CSI-M $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
Level 1	0.3602	0.4616	0.7316	368.22	0.4085	0.5482	0.8389	11.54
Level 2	0.3638	0.4668	0.7284	371.34	0.4023	0.5417	0.8415	11.05
Level 3	0.3555	0.4553	0.7287	370.95	0.3991	0.5379	0.8379	12.06
Level4	0.3569	0.4571	0.7286	367.26	0.3999	0.5388	0.8385	11.33

	Shanghai				CIKM			
	CSI-M $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
Level 1	0.4320	0.5732	0.8156	30.13	0.3131	0.4039	0.6645	38.70
Level 2	0.4307	0.5719	0.8164	30.07	0.3303	0.4266	0.6696	38.64
Level 3	0.4345	0.5768	0.8128	29.57	0.3175	0.4109	0.6705	38.26
Level4	0.4183	0.5585	0.8114	30.82	0.3208	0.4161	0.6686	38.94

F Additional Hyper Parameters Details

Table 8 reports the hyperparameter search space and the best configuration found per dataset via validation CSI-M. The search covers the wavelet family and decomposition level, the spatial convolution kernel size k in the SRST Block spatial branch, the number of groups N and group normalization groups ρ in the STR module, the Gabor frequency multipliers f_l and f_h for the low- and high-frequency FAT Blocks respectively, and the Gabor weight initialization scales W_l and W_h .

The selected configurations are consistent across datasets with two notable exceptions. First, the wavelet level is set to $J = 2$ for SEVIR and $J = 3$ for Shanghai both datasets with richer multi-scale precipitation structure, while $J = 1$ suffices for MeteoNet and $J = 2$ for CIKM. Second, the high-frequency Gabor multiplier f_h is uniformly large (4.0) across all datasets except SEVIR (4.0), reflecting the consistently turbulent character of high-frequency subbands regardless of climatology. The weight initialization scale $W_h = 0.25$ for CIKM (versus 1.0 elsewhere) reflects the comparatively lower precipitation intensity in the Guangdong region, where a smaller initial weight scale stabilizes early training. All other hyperparameters are stable across datasets, confirming that DAWN-Cast does not require extensive per-dataset tuning beyond the wavelet configuration.

G Statistical Analysis

To validate the robustness of our reported findings, we perform statistical analyses over five independent runs using distinct random seeds. Table 9 presents the mean and standard deviation of the main evaluation metrics.

Table 8: Hyperparameter search space and best values found per dataset. **Bold** indicates the selected value used in the final model.

Hyperparameter	Search Range	SEVIR	MeteoNet	Shanghai	CIKM
Wavelet	Haar, db4, db6	db6	db6	db6	db4
Wavelet-level (J)	1,2,3,4	2	1	3	2
Spatial Kernel K	{3, 7, 9}	3	3	3	7
SRST(Spectral Branch) N	{1, 2, 4, 8}	4	4	4	1
SRST (Spectral Branch) ρ	{1, 2, 3, 4, 8}	4	4	3	1
Low-Freq Multiplier f_ℓ	{0.1, 0.2, ..., 2.0, ..., 6.0}	0.1	0.1	4.0	0.1
High-Freq Multiplier f_h	{0.1, 0.2, ..., 2.0, ..., 6.0}	4.0	4.0	4.0	0.1
Weight initialization W_ℓ	{0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0}	0.1	0.1	0.1	0.1
Weight initialization W_h	{0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0}	1.0	1.0	1.0	0.25

Table 9: Statistical analysis of DAWN Cast over five different runs

Dataset	CSI-M $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
CIKM	0.3224 ± 0.0064	0.4175 ± 0.0069	0.6713 ± 0.0042	38.32 ± 0.2605
Meteonet	0.4064 ± 0.0040	0.5459 ± 0.0051	0.8448 ± 0.0021	11.42 ± 0.5101

H Additional Visualization Results

Figure 6 visualizes the progressive improvement in prediction quality on a representative MeteoNet event as components are added incrementally, corresponding to the ablation results of Table 2.

Only WGTM. With only the Wavelet Guided Temporal Modelling block, the model produces predictions that capture the overall large-scale motion and spatial extent of the precipitation system. The general storm trajectory and dominant convective structure are reasonably preserved across the prediction horizon. However, the predictions appear spatially coarse and lack fine-grained internal structure, with precipitation boundaries that are diffuse and high-intensity cores that are underestimated at longer lead times (T_{16} , T_{20}). This reflects the WGTM’s role as a temporal modelling block, it correctly forecasts *when* and *where* precipitation evolves, but without subsequent spatial refinement the output remains structurally incomplete.

Adding the Spectral Branch. Incorporating the Spectral Temporal Refinement (STR) branch of the SRST Block produces a notable improvement in global structure. The large-scale precipitation organization becomes sharper and more coherent, and the predicted storm morphology aligns more closely with the ground truth at all lead times. This is consistent with the global receptive field of the Fourier-domain processing in the STR branch, which suppresses spatially distributed artifacts and refines the overall motion structure of the predicted field.

Adding the Spatial Branch (Full model). The addition of the spatial branch, depthwise convolutions for localized fine-scale refinement produces the most visually striking improvement. Fine-grained precipitation boundaries become sharper, internal intensity gradients within storm cells are recovered, and high-intensity cores (green and yellow regions) are more accurately reproduced at longer lead times. The full model prediction closely matches the ground truth in both spatial structure and intensity distribution, confirming that the spectral and spatial branches provide complementary and non-redundant refinement: the former corrects global structure, while the latter recovers local detail that global Fourier processing cannot provide.

We present additional qualitative results on the SEVIR, Meteonet, Shanghai, and CIKM datasets to complement the examples shown in the main paper.

Fig. 7 and Fig. 8 show additional qualitative results on the SEVIR dataset. Fig. 9 and Fig. 10 present further examples on the Meteonet dataset. Similarly, Fig. 11 and Fig. 12 correspond to the Shanghai dataset, while Fig. 13 and Fig. 14 provide additional results on the CIKM dataset.

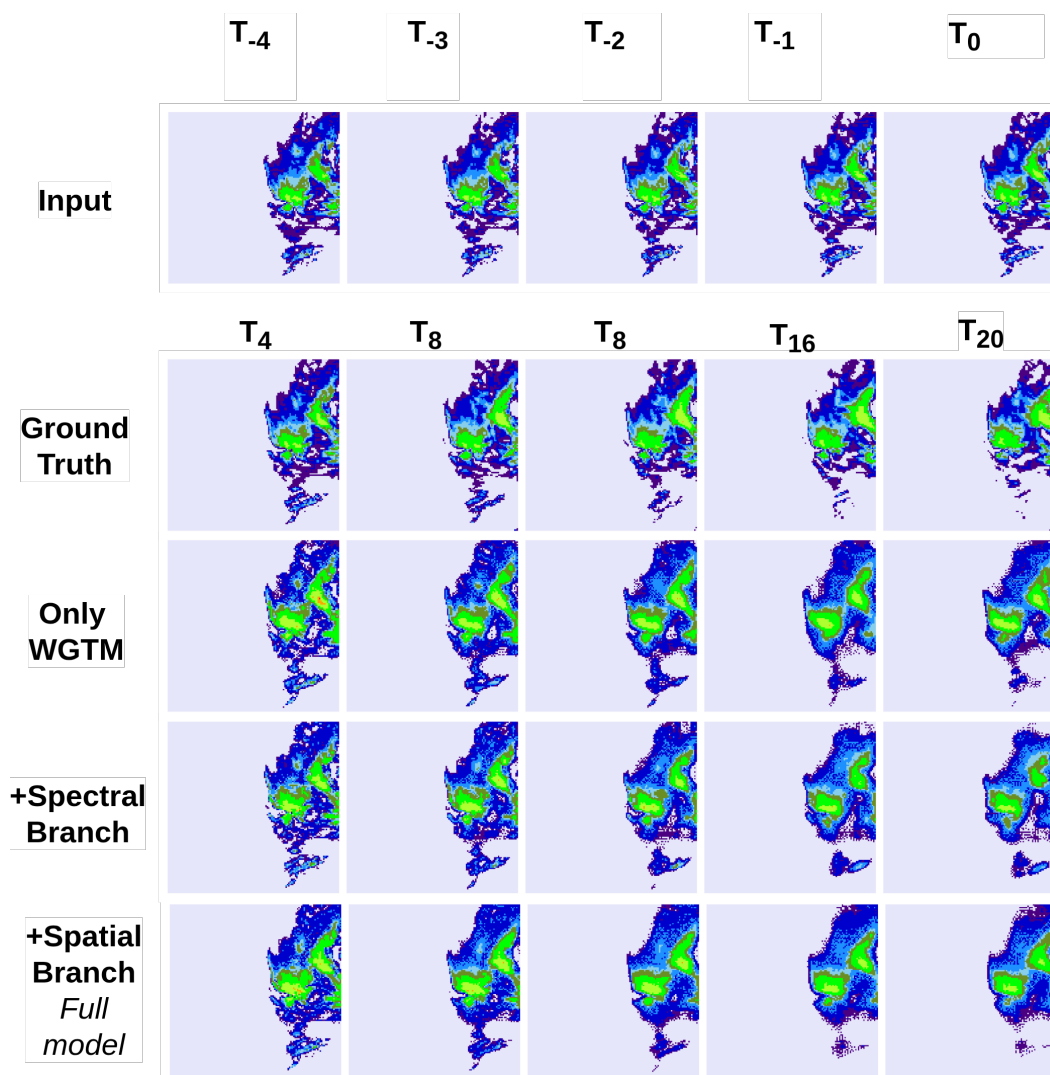


Figure 6: Visualization of the evolution of predictions on incremental component addition.

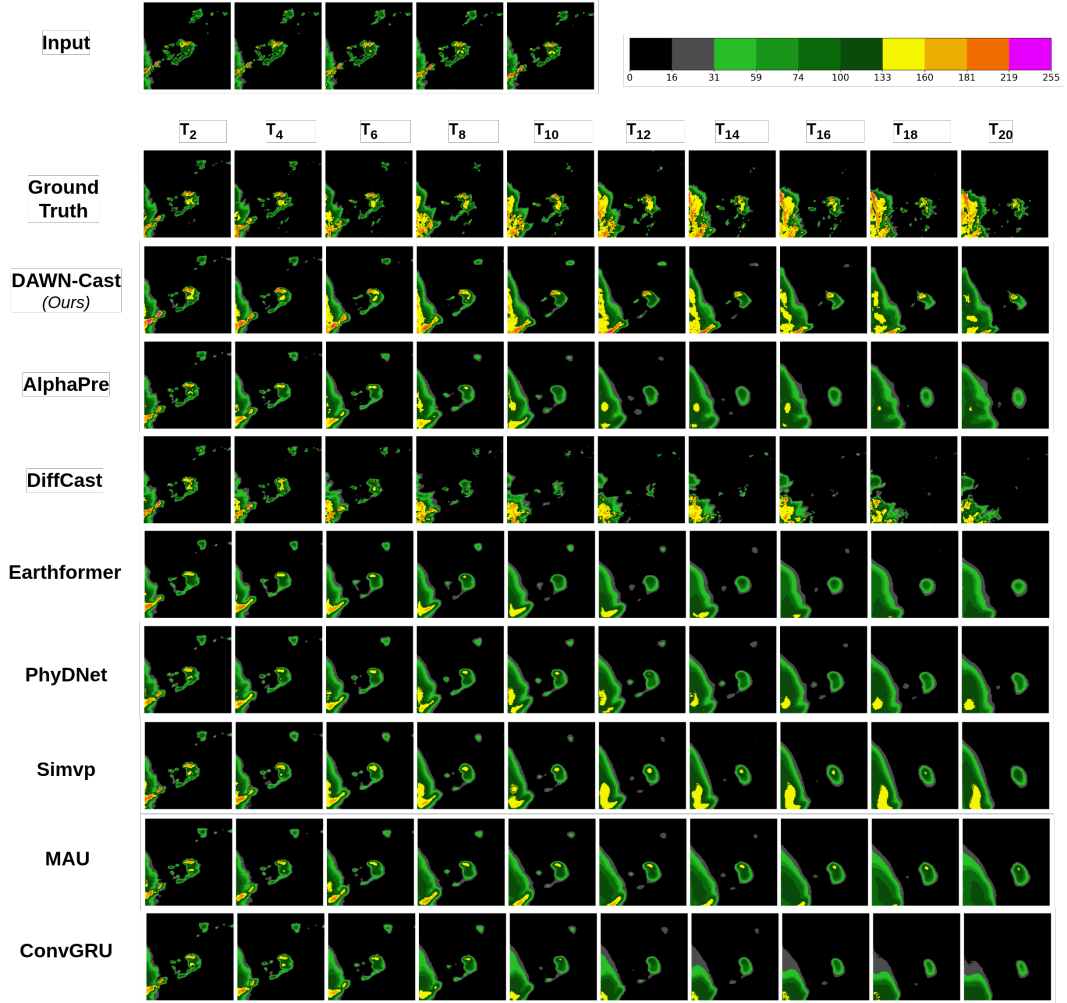


Figure 7: Additional qualitative results on the SEVIR dataset (I).

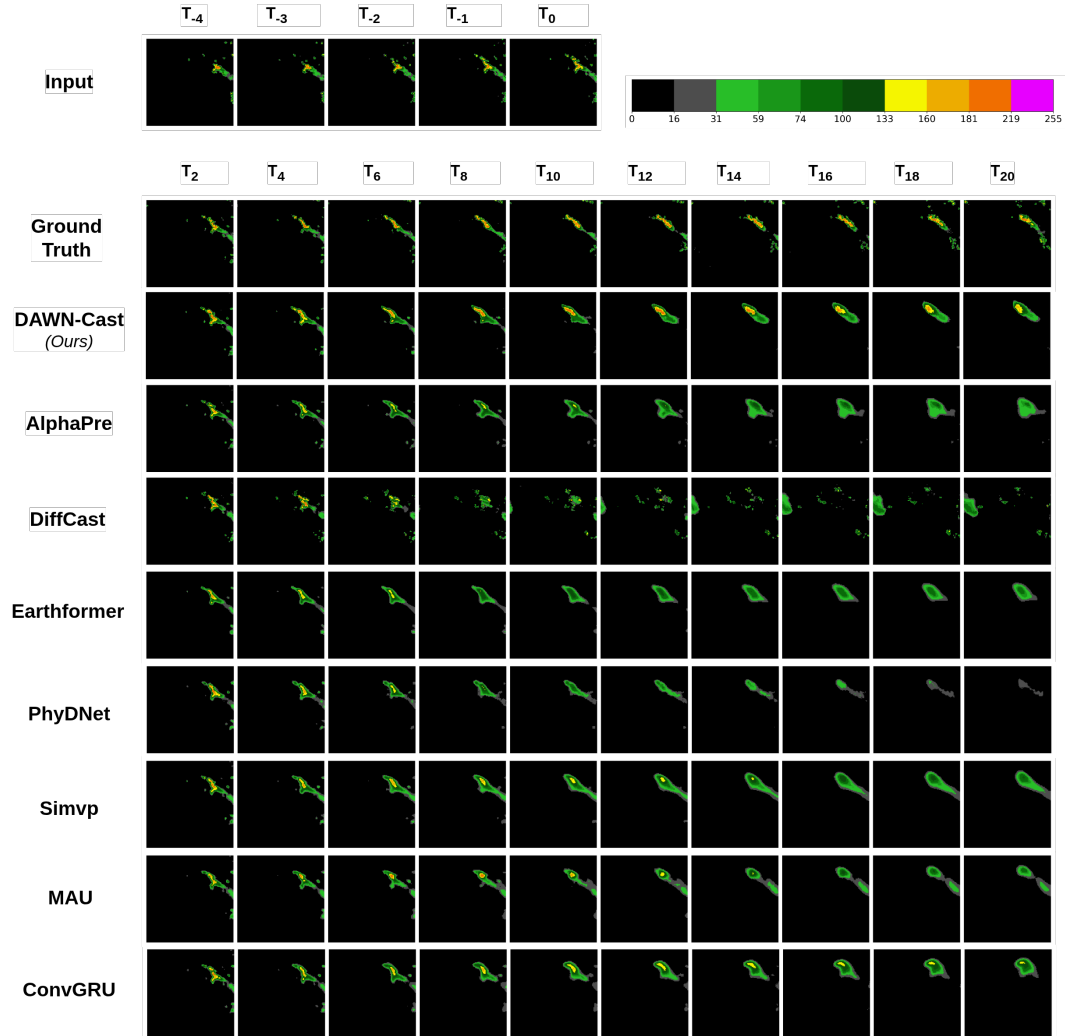


Figure 8: Additional qualitative results on the SEVIR dataset (II).

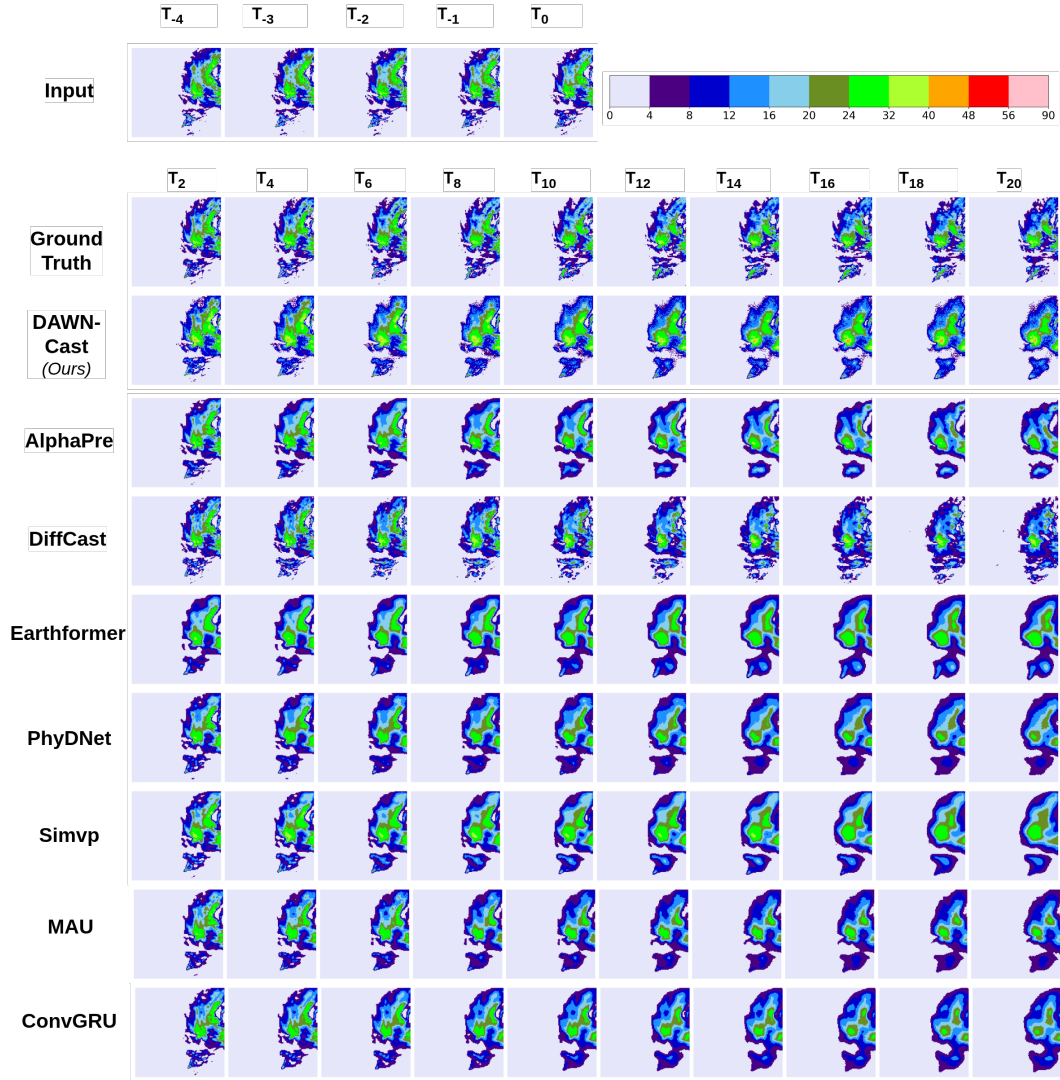


Figure 9: Additional qualitative results on the Meteonet dataset (I).

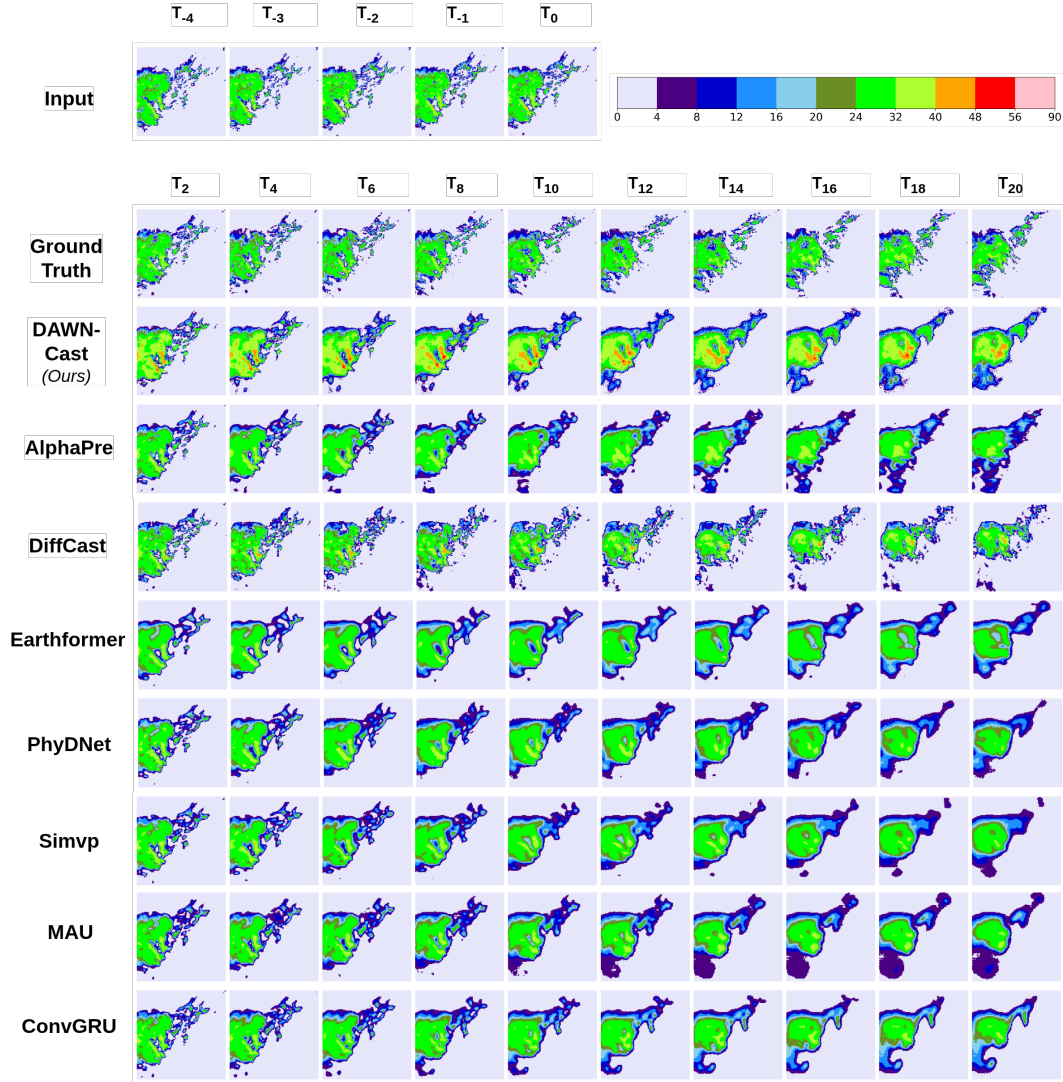


Figure 10: Additional qualitative results on the Meteonet dataset (II).

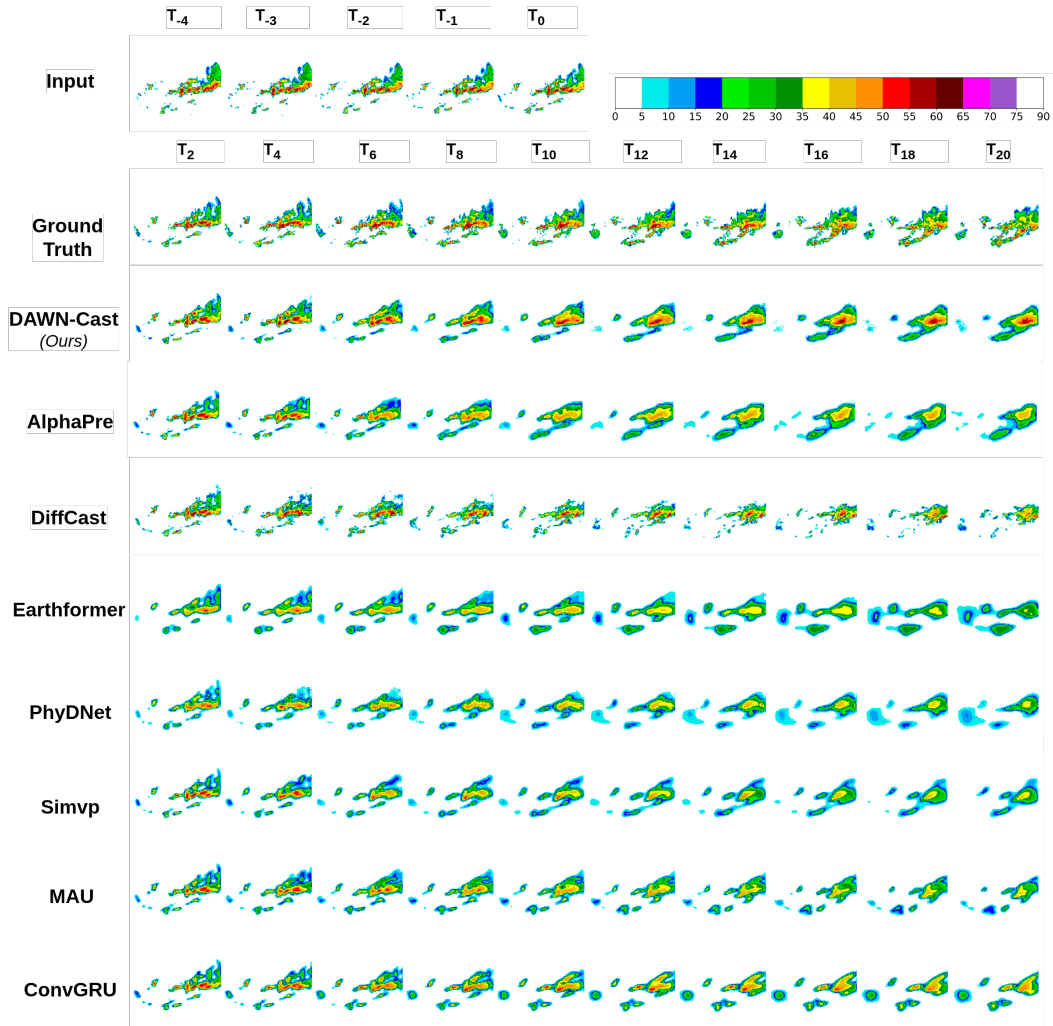


Figure 11: Additional qualitative results on the Shanghai dataset (I).

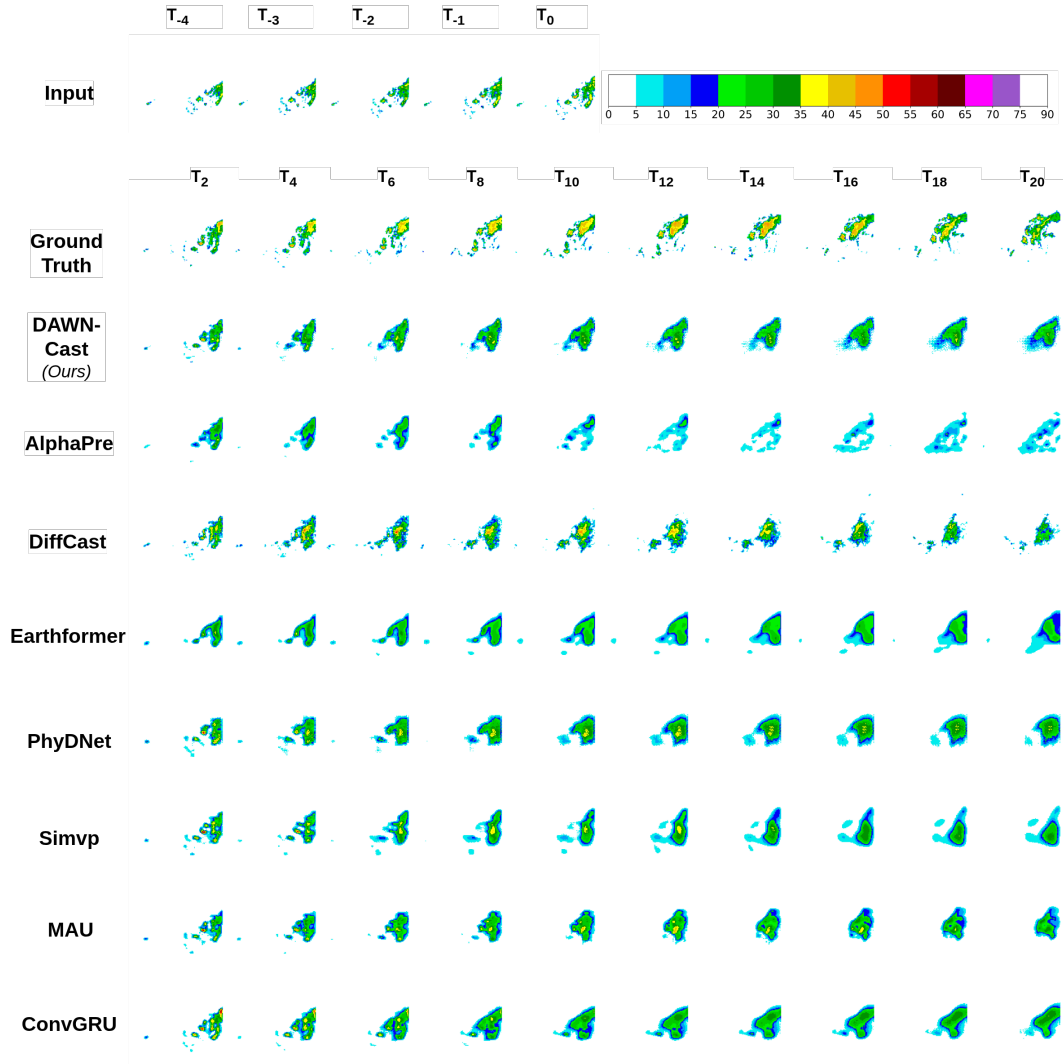


Figure 12: Additional qualitative results on the Shanghai dataset (II).

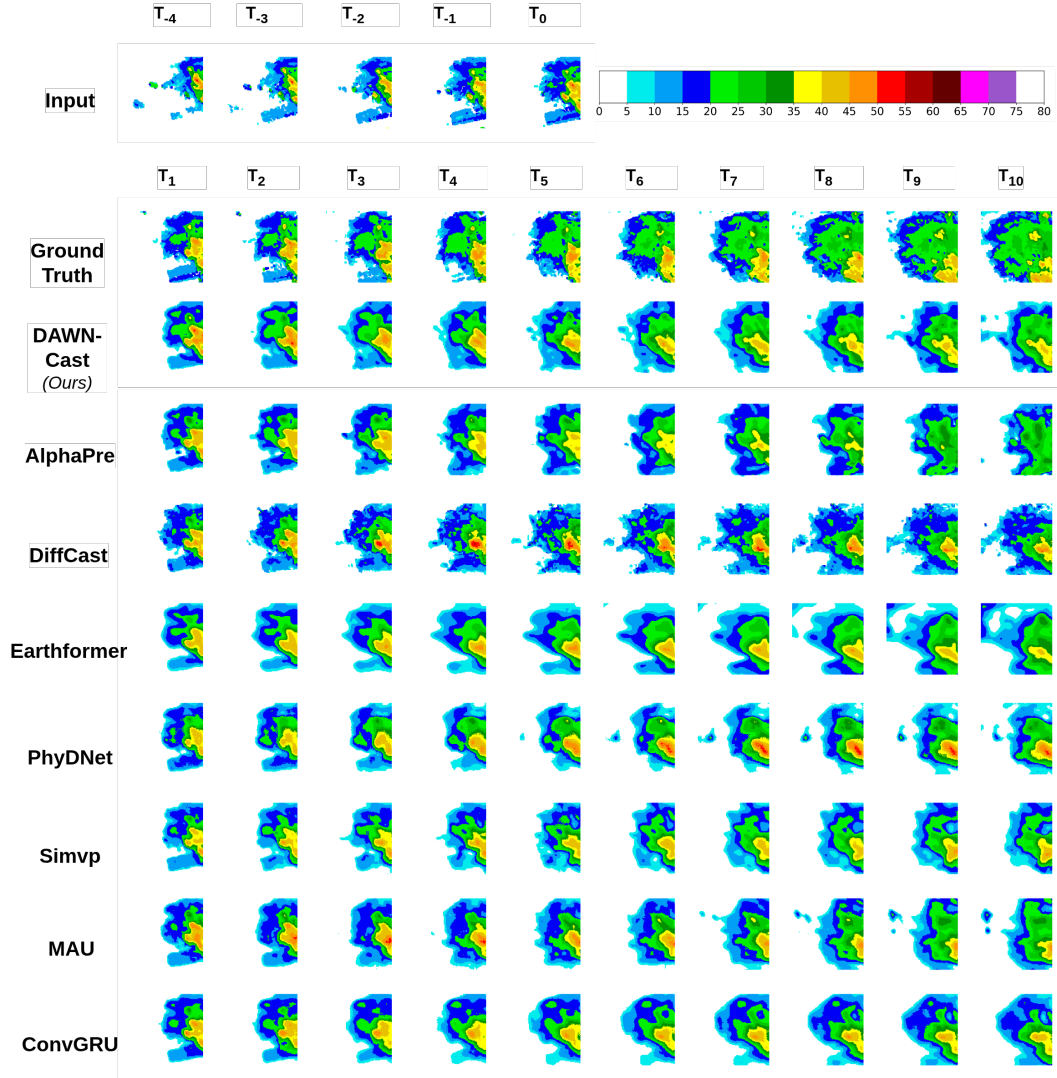


Figure 13: Additional qualitative results on the CIKM dataset (I).

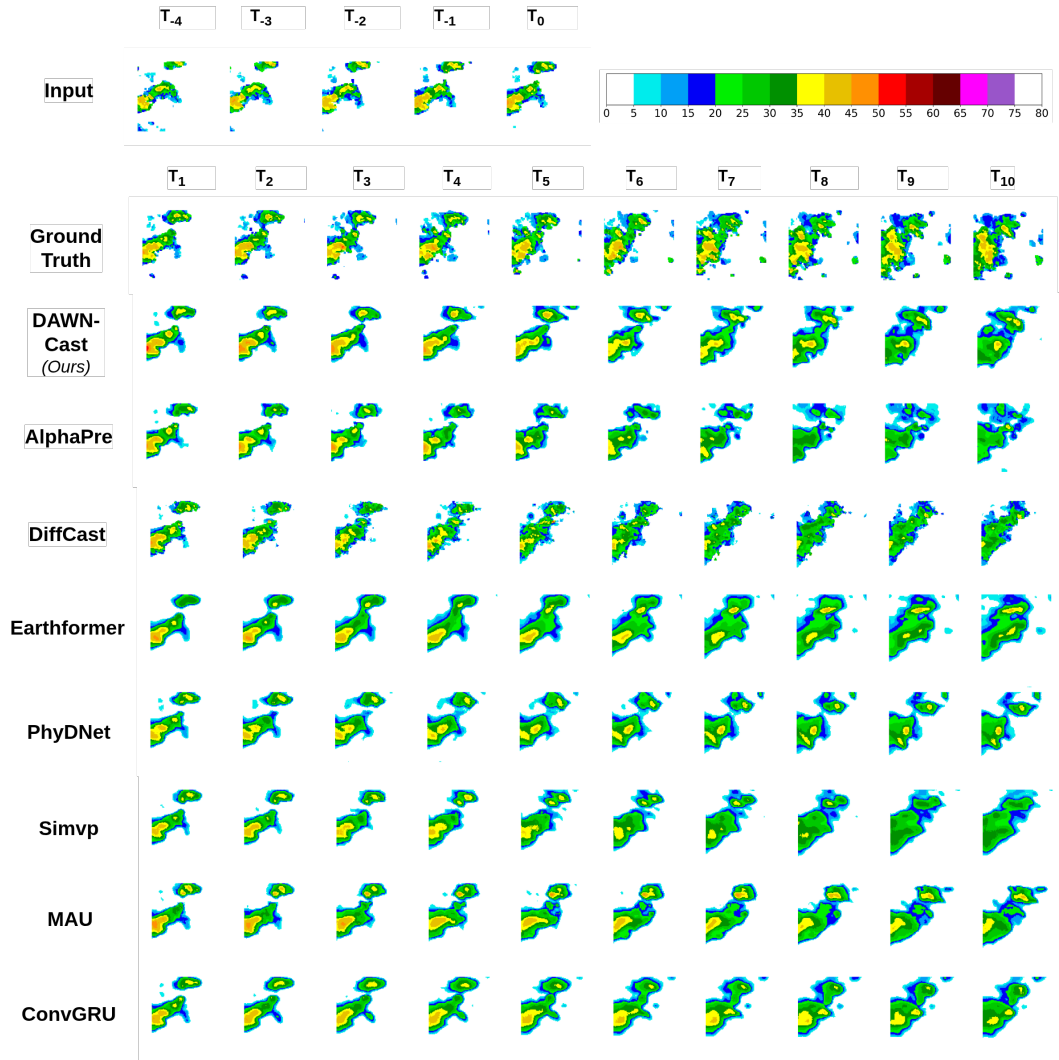


Figure 14: Additional qualitative results on the CIKM dataset (II).

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16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: We have used LLMs only for English language grammar corrections.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.